

View results

Respondent

54

Anonymous

201:11

Time to complete

A. General information about the respondent

1. * Country

Greece



2. * Name of organization(s) or entity(s) responsible for the preparation of the report

General Secretariat for Research and Innovation; Athena Research Center

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

<https://gsri.gov.gr/en/> ; <https://www.athenarc.gr/en>

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

GSRI: Research Funding Organisation; ARC: Research Performing Organisation competent in Open Science.

7. * Full name of officially designated contact person(s) who completed this survey

Maria Christoula; Natalia Manola; Elli Papadopoulou

8. * Position of officially designated contact person(s) who completed this survey

GSRI: Director, International Scientific & Technological Cooperation Directorate; ARC: Research Associates

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

Maria Christoula

11. Brief description of the consultation process established for the preparation of the report

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

☒ Public

☐ Private

☐ Civil Society

☐ Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

☒ Yes

☐ No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

On February 15, 2022, an online event titled "UNESCO: Open Science – Artificial Intelligence. New Horizons, Visions, and Boundaries" was held to present the UNESCO Recommendations on Open Science to the Greek scientific community (<https://elpida.org/%CE%B4%CE%B9%CE%B1%CE%B4%CE%B9%CE%BA%CF%84%CF%85%CE%B1%CE%BA%CE%B7-%CE%B7%CE%BC%CE%B5%CF%81%CE%B9%CE%B4%CE%B1-unesco-%CE%B1%CE%BD%CE%BF%CE%B9%CF%87%CF%84%CE%AE-%CE%B5%CF%80%CE%B9%CF%83%CF%84/>). Additionally, the Recommendations were communicated to events where other Ministries were present such as at the 1st Greek EOSC Tripartite Event in October 2022 (<https://eosc.eu/news/greek-open-science-community-gathered-tripartite-event>), and the 3rd Open Science Symposium and EOSC Tripartite Event in Greece in December 2024 (<https://eosc.eu/news/2024/12/national-tripartite-event-in-greece-highlights-community-as-the-driving-force-for-open-science/>). The latter was held under the joint auspices of the Ministries of Development, of Digital Governance, and of Education, Religious Affairs and Sports.

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

☐ Yes

☒ No

19. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) 16.f, 16.g, 16.h)

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

☒ Yes

☐ No

20. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

Established in February 2022 by 13 leading Greek research, technology, and innovation organisations, the Hellenic Open Science Initiative (HOSI) aims to implement Open Science policies in Greece and support national representation in the European Open Science Cloud (EOSC). Among other things, HOSI's Training Working group has been active in preparing and promoting training material (see: <https://zenodo.org/communities/os-greece/records?q=&l=list&p=1&s=10&sort=newest> & <https://palc30.lib.uoa.gr/%cf%80%cf%81%ce%bf%ce%b3%cf%81%ce%b1%ce%bc%ce%bc%ce%b1-%cf%83%cf%85%ce%bd%ce%b5%ce%b4%cf%81%ce%b9%ce%bf%cf%85-2/>) as well as co-organised the training events on the second day of the 2024 Open Science Symposium. OpenAIRE plays a key role in promoting Open Science in Greece through its National Open Access Desk (NOAD), hosted by Athena Research Center, providing training, policy support, and infrastructure development. The Greek NOAD has been promoting Open Science and spreading awareness on new developments and services for implementation since 2009 and collaboratively organising webinars with the Cypriot NOAD. Some statistics can be found here: <https://zenodo.org/records/14944907> Additionally, individual HOSI members are active in European projects that support the adoption and implementation of Open Science which facilitates knowledge transfer to their institutions and the country at large. For more statistics and progress on how Greece and other countries perform in EOSC can be found at the EOSC Observatory: <https://eoscobservatory.eosc-portal.eu/home>.

21. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? (*ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h*)

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

☒ Yes

☐ No

22. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

A new version of the National Open Science Plan (<https://zenodo.org/records/14808290>), currently under review by Greek authorities (Ministry of Digital Governance, MERAS, and GSRI), lays the groundwork for a comprehensive Open Science Strategy. The plan covers key areas such as open access, research data, software development, and e-infrastructures, detailing 62 actions that include infrastructure development, education policies, funding mechanisms, and legislative measures. The Hellenic Open Science Initiative (HOSI) plays a central role in shaping and implementing this strategy, bringing together researchers, library organisations, and technology providers to foster an open and collaborative research environment. Concrete steps are being taken to integrate Open Science principles into publicly funded research. Efforts are underway to include Open Science requirements in funding calls by the GSRI and the Hellenic Foundation for Research & Innovation (HFRI), with a particular emphasis on ensuring Open Access to scientific publications. These actions align with the values and guiding principles of Open Science, including quality and integrity, equity and fairness, transparency, responsibility, collaboration, and sustainability, ensuring that Open Science becomes a core component of Greece's research ecosystem by 2025.

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

23. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

☒ Yes

☐ No

24. 2.1 cont. **If yes**

- please share a machine-readable PDF file of the policy using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

Title of the Policy: Law 4310/2014, titled "Research, Technological Development and Innovation and Other Provisions." Authority in Charge of Development: The policy was developed under the auspices of the Hellenic Ministry of Education, Research, and Religious Affairs. Since 2019, due to restructuring of the public administration bodies, amendments have been undertaken by the Ministry of Investments. GSRI has been at the forefront of these activities from the beginning (before 2019 named General Secretariat for Research and Technology – GSRT). Entities Involved in Development: The formulation of Law 4310/2014 involved multiple stakeholders, including: • General Secretariat for Research and Innovation: Provided strategic direction and oversight. • Greek Atomic Energy Commission (EEAE): Contributed expertise, particularly for sections related to nuclear energy and radiation protection. • Research and Academic Institutions: Offered insights and feedback during the drafting process. • National Council for Research and Technology (NCRT-ESET) Process of Development: The development of Law 4310/2014 followed a structured process, starting with consultations involving scientific communities, research institutions, and industry stakeholders. This was followed by drafting a policy framework aligned with national research and innovation goals. The draft then underwent review and revision, incorporating feedback from public consultations and expert assessments. Finally, the refined policy was submitted for legislative approval and enacted by the Hellenic Parliament. (A draft analysis in Greek can be found at: <https://zenodo.org/records/4502176>) Year of Adoption: The law was adopted on December 8, 2014. Year of Last Update: The most recent amendment occurred in 2022, with the enactment of Law 4982/2022, which introduced updates to specific provisions. Key Areas/Sections of the Policy: • Research and Innovation Framework: Establishes the national strategy for research, technological development, and innovation. • Institutional Structure: Defines the roles and responsibilities of research organisations and oversight bodies. • Funding Mechanisms: Outlines financial support structures for research and innovation activities. • Pre-Commercial Procurement (PCP): Introduces the concept of PCP to promote innovation through public sector demand. • Nuclear Energy and Radiation Protection: Details regulations concerning the management of nuclear energy and assigns regulatory authority to the EEAE. Webpage Links: • Full Text of Law 4310/2014: <http://www.gsrt.gr/Legislation/Files/LawFiles136/%CE%9D%CE%B5%CE%BD%CE%BF%CF%80%CE%BF%CE%B9%CE%B7%CE%BC%CE%AD%CE%BD%CE%BF%CF%82.pdf> (in Greek) • Explanatory Memorandum: https://www.minedu.gov.gr/publications/docs2015/Explanatory_Memorandum_to_Law_4310_2014.pdf (in English)

25. 2.1 cont. **If yes:**

- does it include any of the following open science elements:

- ☒ open access to scientific knowledge
- ☐ open science infrastructure
- ☐ open engagement of societal actors
- ☐ open dialogue with other knowledge systems

26. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (*ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h*)

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ☐ Yes
- ☐ Partly
- ☒ No

27. 2.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.

Greece faces a unique challenge in Open Science governance due to the distribution of responsibilities across three ministries. As highlighted during the 3rd Open Science Symposium, this structure requires clear guidelines and enhanced coordination, particularly in the ongoing update of the Digital Transformation Bible. Strengthening inter-ministerial collaboration is crucial to avoiding past gaps, such as the limited involvement of the Ministry of Development and the Ministry of Education, Religious Affairs, and Sports in Open Science discussions due to their exclusion from relevant processes.

28. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

- ☐ Yes
- ☐ Under development
- ☒ No

29. 2.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

The GSRI is actively working to strengthen coordination and secure funding for the successful development and implementation of Open Science policies in Greece. Particularly, efforts are underway to enable the Hellenic Open Science Initiative (HOSI) to implement concrete actions from the national plan. However, these efforts are still at a preliminary stage.

30. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)

- ☐ Yes
- ☒ Under development
- ☐ No

31. 2.4 cont. **If yes or under development**, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

The GSRI is actively working to strengthen coordination and secure funding that will enable the adoption of the National Open Science Plan as a national strategy.

32. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 17.e, 17.g)

- ☒ Yes
- ☐ Under development
- ☐ No

33. 2.5 cont. **If yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

To view the elements of open science,

see: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=9>

Five universities in Greece have already adopted institutional Open Access (OA) policies (<https://roarmap.eprints.org/view/country/300.html>), while all universities maintain self-archiving policies linked to their institutional repositories, ensuring the preservation of Open Access publications. The HOSI plays a pivotal role in harmonizing the adoption of Open Science policies across institutions. As a result, more universities are expressing interest in joining HOSI, further accelerating institutional adoption. This momentum is also reinforced by the 71st Rectors' Conference, where Open Access and Open Science were officially recognized, driving a coordinated and systematic approach to policy implementation in Greek universities.

34. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? (*ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c*)

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

- ☐ Yes
- ☒ Under development
- ☐ No

35. 2.6 cont. **If yes or under development**, please provide details and links as relevant.

While there is no dedicated national financial strategy, the Hellenic Academic Libraries Link (HEAL-Link) serves as the national body responsible for negotiating agreements with scientific publishers, ensuring access to scientific information for the Greek research community. HEAL-Link also facilitates Open Access agreements that waive publication fees. These agreements are primarily funded through national resources from the Ministry, but individual institutions may also allocate local budgets for specific electronic resource acquisitions that meet their needs. More details are available at: HEAL-Link Open Access Agreements. In addition, some funding for electronic infrastructures has been allocated to GRNET, the National Infrastructures for Research and Technology provider in Greece, for enhancing capacity to support research.

36. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

☒ Yes

☐ No

37. 2.7 cont. **If yes**, please provide details and links as relevant.

There are recent efforts to include Open Science requirements in calls for projects of the Hellenic Foundation for Research & Innovation (HFRI) and EREVNO KAINOTOMO of GSRI, with a special focus on Open Access to publications.

38. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

☐ Yes

☒ No

39. 2.8 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

There is strong engagement of the policy community to adopt Open Science in the public sector. Once Greece has this foundation at national level, more actions towards PPPs are expected to be planned.

40. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

- ☐ Yes
- ☒ Under development
- ☐ No

41. 2.9 cont. **If yes or under development**, please provide details and links as relevant.

Greece is actively contributing to the development of the EOSC Observatory, which tracks national progress on Open Science policy implementation, funding, infrastructures, skills, and engagement with the European Open Science Cloud (EOSC). While this is not a national monitoring mechanism per se, the GSRI is working towards a structured approach to monitoring Open Science activities in Greece by exploring best practices, such as the Irish Open Science Monitor developed by OpenAIRE, which provides a structured framework for tracking Open Science adoption. Furthermore, HEAL-Link has been monitoring Open Access to publications through surveys and reports (available here: <https://scholarly.heal-link.gr/monitoring/>) and is currently developing an OpenAIRE Monitor for its academic community.

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

42. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

As of December 2024, the Organisation for Economic Co-operation and Development (OECD) reports that Greece's gross domestic expenditure on research and development (R&D) remains at 1.3% of its Gross Domestic Product (GDP). See: https://www.oecd.org/content/dam/oecd/en/publications/reports/2024/12/oecd-economic-surveys-greece-2024_26a6b3e5/a35a56b6-en.pdf

43. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

As of early 2024, approximately 86.9% of Greek households have internet access. Efforts are ongoing to enhance digital infrastructure, as outlined in the National Broadband Plan 2021-2027 (<https://digital-strategy.ec.europa.eu/en/policies/broadband-greece>), which aims to promote the use of fixed very high-capacity and 5G networks as catalysts for the country's digital transformation.

44. 3.3 Are there national research and education networks (NRENs)[i] active in the country? (ref.: (iii) 18.c)

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

☒ Yes

☐ No

45. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

GRNET is the NREN for Greece providing advanced network, cloud computing, and High-Performance Computing (HPC) services to the country's educational, academic, and research communities.

46. 3.4 Are there national or institutional open science infrastructures[i] in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

- ☐ Yes
- ☒ Under development
- ☐ No

47. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- ☒ federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- ☐ community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society
- ☐ platforms for exchanges and co-creation of knowledge between scientists and society
- ☒ community-based monitoring and information systems to complement national, regional and global data and information systems

48. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

Greece has several national and institutional Open Science infrastructures that support preservation, management, and open access to scientific information. ATHENA RC and GRNET have developed the HELIX (Hellenic Data Service) infrastructure to address long-term data preservation needs. HELIX, recognized as one of the core national research infrastructures in Greece's National Roadmap for Research Infrastructures, is designed according to Open Science and Open Source best practices, ensuring compliance with OpenAIRE guidelines for interoperability and supporting FAIR data principles. HARDMIN, a twin infrastructure to HELIX developed by ATHENA RC, is managed by HEAL-Link and serves Greek universities by providing interconnected services for research data storage and reuse. In addition, HEAL-Link maintains a literature repositories network, enhancing access to scientific information. ARGOS, OpenAIRE's machine-actionable Data Management Planning (DMP) service, has been deployed in HEAL-Link's infrastructure, integrating with HARDMIN and academic services to automate and validate the DMP process while promoting early-stage data sharing. Beyond institutional repositories, Greek Nodes of Research Infrastructures, such as SODANET and ELIXIR-GR, contribute by offering services catalogues and developing best practices for Research Data Management (RDM) and FAIR data implementation in their designated domains. Overall, while significant progress has been made, continued funding and policy support are essential to enhance and sustain these Open Science infrastructures.

49. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

☒ Yes

☐ No

50. 3.5. cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

Name of the Infrastructure: OpenAIRE (Open Access Infrastructure for Research in Europe)
Date of Establishment: 2009 Language: English (primary), with multilingual support for national services
Target Groups & Beneficiaries: • Researchers, research institutions, universities • Funders and policymakers • Librarians, data stewards, research support staff • Repository managers • Citizen science communities • Publishers and infrastructure providers
Governance / Community Agreements: OpenAIRE is governed as a non-profit organization (OpenAIRE AMKE), with over 50 partners across Europe, including research institutions, libraries, and e-infrastructure providers. It operates on the principles of collaboration, inclusivity, and transparency, engaging stakeholders through its human and technical infrastructure. The prior is realised by maintaining and mobilizing National Open Access Desks (NOADs), which connect OpenAIRE with national research communities, while the latter is possible through the OpenAIRE Graph, the largest Open Science knowledge graph in Europe.
Scope & Key Areas: OpenAIRE is a horizontal infrastructure aimed at being used by all domains. At the heart of all its services and operations is the OpenAIRE Graph, however the three main areas it focuses on are: 1. Policies: to facilitate adoption and compliance with Open Access and Open Science mandates as well enable data-driven insights into Open Science adoption through assessments. 2. Training: to promote community engagement, build capacities and foster open innovation 3. Services: to equip the scientific community with the necessary tools and standards for FAIR data management and stewardship. Accessibility: OpenAIRE services are accessible globally, serving the European research ecosystem while integrating with international Open Science efforts. Through collaborations with institutions and policymakers, OpenAIRE ensures that its services, policies, and guidelines are widely applicable and aligned with Horizon Europe and EOSC mandates.

51. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

☒ Yes

☐ No

52. 3.6 cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

Both GRNET and OpenAIRE support collaborations with the South East and North West regions through its networks. Below we provide more information on GRNET. Name of the Infrastructure: Greek Research and Technology Network (GRNET) Date of Establishment: 1998 Language: Greek and English Target Groups and Beneficiaries: • Educational Institutions: Universities, Technological Education Institutes, and over 9,500 schools across Greece. • Research Centers: Providing infrastructure and services to facilitate advanced research activities. • Healthcare Sector: Public hospitals benefit from GRNET's services. • Cultural Institutions: Museums and libraries are also among the beneficiaries. Governance/Community Agreements: GRNET operates under the auspices of the Greek government, specifically the Ministry of Digital Governance. It collaborates with various national and international organizations to develop and manage its infrastructure and services. As the coordinator of the Authentication and Authorization Infrastructure (AAI) Federation in Greece, GRNET manages processes related to the integration and withdrawal of federation members, ensuring compliance with established policies. Scope: • Research Areas: GRNET supports a wide range of scientific research areas by providing the necessary digital infrastructure, including cloud computing and HPC resources. • Elements of Open Science: GRNET is instrumental in promoting Open Science practices in Greece, is a founding member of HOSI and provides technical infrastructure to support FAIR (Findable, Accessible, Interoperable, and Reusable) data management practices. • Stage of Research: GRNET's services cater to various stages of research, from initial data collection and processing to advanced computational analysis and long-term data storage. • Accessibility: GRNET's infrastructure and services are accessible to all relevant users within Greece.

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY

AND CAPACITY BUILDING FOR OPEN SCIENCE

53. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)



Yes



No

54. 4.1 cont. **If yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

There have been multiple bottom-up initiatives aimed at upskilling researchers, academic librarians, and research support staff with Open Science expertise. Both HEAL-Link and Athena Research Center serving as the OpenAIRE NOAD play a crucial role in training and capacity building for Open Science in Greece. Through a series of webinars and training events, they provide practical guidance on Research Data Management (RDM), data management planning, FAIR principles, repository use, interoperability, and compliance with funding requirements, including Horizon Europe mandates. HEAL-Link's Scholarly Communication Unit webinars focus on supporting researchers, librarians, and institutional policymakers in adopting Open Science practices while OpenAIRE complements these efforts by engaging research institutions, funders, and service providers across Greece and Europe, ensuring the seamless integration of Open Science principles into the research ecosystem. In 2022, the RDM Training Programme for Academic Librarians was offered by OpenAIRE NOAD in Greece, focused on FAIR data principles, data curation, and best practices for supporting researchers in managing research data. This initiative strengthened the role of academic libraries in Open Science by providing tailored guidance and practical training. More lately, the HOSI Training Working Group coordinated systematic Open Science training activities across Greek institutions-HOSI members. The group has produced training resources and will be organizing workshops, webinars, and knowledge-sharing sessions to support the adoption of Open Science in 2025. Lastly, HOSI members participating in European Open Science projects might also perform training activities of sorts in 2025.

55. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

☐ Yes

☒ No

56. 4.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

We are currently planning such activities.

57. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

☐ Yes

☒ No

58. 4.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

There are scattered efforts from HOSI members active in European Open Science projects to create such courses embedded in University curricula, eg RAISE Suite which is a forthcoming EOSC project will develop a course and roll it out at the Aristotle University of Thessaloniki (AUTH).

59. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

☒ Yes

☐ Partly

☐ No

60. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

☒ Yes

☐ No

61. 4.5 cont. **If yes**, please provide more information and links as relevant.

There is an annual event taking place across regions in Greece titled "researchers' night" where citizens and students get the opportunity to learn the science behind everyday products, natural phenomena, the technology, etc. In addition, Athena Research Center and HOSI have been extending invitations to national Open Science Symposia to gather all Open Science stakeholders together (last one being the 3rd Open Science Symposium in Dec 2024).

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

62. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c, 20i)

☒ Yes

☐ No

63. 5.1 cont. **If yes**, please provide more information and links as relevant.

While significant progress has been made by individual organisations in the context of their signatory and participation to CoARA, challenges remain in embedding these principles into formal evaluation frameworks. More specifically: • ATHENA Research Center (ARC) is the only Greek organisation to join CoARA and is currently working towards updating its policies to adopt responsible research assessment (RRA) practices. • The National and Kapodistrian University of Athens (NKUA) closely monitors CoARA developments as a signatory, preparing for potential alignment with Open Science-friendly evaluation practices. • A national CoARA Chapter is currently being developed, led by ARC and supported by HOSI, to coordinate efforts and foster institutional engagement in research assessment reform. • The Hellenic Authority for Higher Education (HAHE) is aware of Open Science principles, but these are not yet integrated into its monitoring and evaluation processes.

64. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

☐ Yes☒ No65. 5.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

66. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

☐ Yes☒ No

67. 5.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

68. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

☐ Yes

☒ No

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES

OF THE SCIENTIFIC PROCESS

69. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (*ref.: (vi) 21*)

- ☐ YES, at the national level
- ☒ YES, at the institutional level
- ☐ YES, at both the national and institutional levels
- ☐ No

70. 6.1 cont. **If yes**, please indicate which of the recommended actions they address:

- ☒ promotion of pre-prints
- ☒ exploration of open peer-review practices
- ☐ valuing and sharing of negative scientific results and associated data
- ☒ participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- ☐ development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas
- ☒ promoting open innovation practices
- ☒ pre-registration

71. 6.1. cont. **If yes**, please provide more information and links as relevant

• HEAL-Link's Journal Platform for Open Peer Review and Preprints: a journal platform based on Open Research Europe that embeds open peer review processes and encourages the early sharing of research outputs through preprints. (<https://f1000research.com/heallink>) • Citizen Science Projects and Organizations: The OpenAIRE Citizen Science activities, which support engagement between researchers and the public. Various Greek research institutions and NGOs running citizen-driven environmental, health, and cultural heritage projects, involving non-academic contributors in data collection, analysis, and co-creation of knowledge. • Pre-registration and Open Science Teams in Universities: Some universities, such as the National and Kapodistrian University of Athens (EKPA), are forming pre-registration teams that encourage students and early career researchers to pre-register study designs and hypotheses before data collection. • Promotion of Open Innovation: Public-private partnerships and open innovation hubs fostering technology transfer and co-creation. National initiatives aligned with Horizon Europe, supporting innovation ecosystems that leverage open data, open-source software, and FAIR principles to drive scientific and economic advancements.

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

72. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

Greece facilitates international scientific collaborations on Open Science, as outlined in the 2021 Recommendation on Open Science (vii) 22.a, primarily through its involvement in the Research Data Alliance (RDA) and the international outreach of ESFRI (European Strategy Forum on Research Infrastructures). HOSI members actively contribute to RDA working groups and interest groups (notably CERTH and Athena Research Center have been leading at least 5 groups), helping to shape global best practices in FAIR data and services, reproducibility, software and research data management (RDM). These collaborations and outputs are then communicated to HOSI General Assembly ensuring alignment with global Open Science strategies. At the same time, Greece's participation in ESFRI RIs extends its Open Science outreach across multiple disciplines. Through ELIXIR-GR, Greece supports open bioinformatics data sharing, while DARIAH-GR & CLARIN:EL enhance digital research in humanities and linguistics. CESSDA-GR facilitates cross-border social science data sharing, and EPOS-GR advances the open accessibility and interoperability of geological and seismological data. These collaborations reinforce Greece's commitment to Open Science by integrating its research infrastructures into the broader European and international ecosystem.

73. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

☐ Yes

☒ No

74. 7.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

75. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

☐ Yes

☒ No

76. 7.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Part 8. GENERAL CONSIDERATIONS

77. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

☐ Yes

☒ No

78. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org